Advertising, Personalization, and Search Costs

Evidence from an Online Field Experiment

Cristiana Firullo[1] Cristobal Cheyre[1] Alessandro Acquisti[2]

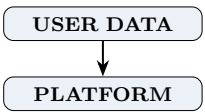
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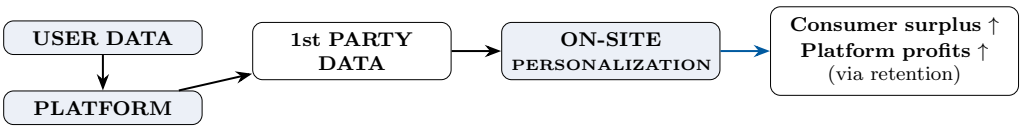
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From platform data to personalization vs ads

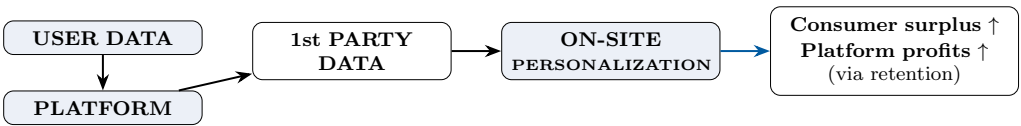


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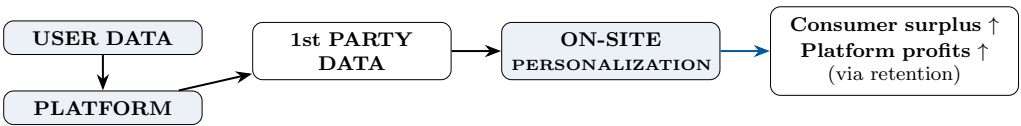
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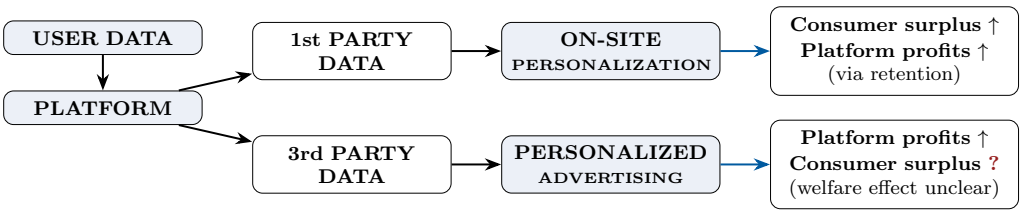
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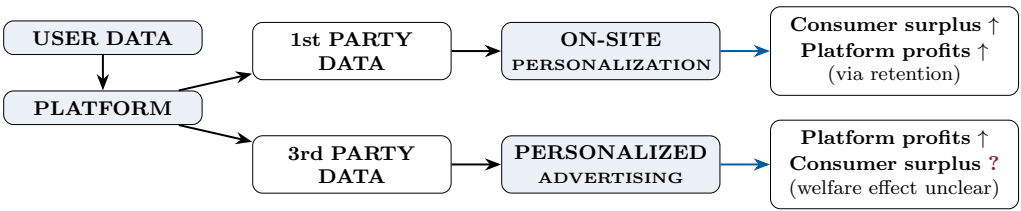
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  - **personalization** ⇒ higher match quality ⇒ retention ⇒ platforms' profits.
  - platforms' incentives are aligned with a **reduction of search costs**.
- However, to maximize **ad revenues**, you might **increase search costs**.
  - Example: personalized ads might distract a user from a specific purchase goal.

This paper

Research questions

- Does **personalized advertising** influence **search time**?
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Empirical setting

- Pre-registered RCT with three arms: **personalized ads** (targeted), **contextual ads only** (untargeted), or **no ads**.
- Full browsing clickstream + rendered HTML content.
- We build a novel approach to study the set of products **seen**, **evaluated**, and **purchased** at the participant level.

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Findings

- Targeted ads raise time per product +13%
- No change in products evaluated, ratings, or purchases ⇒ **friction, not information**

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- We observe *every* page, dwell time, and the products on each search page.

From a search page to the data

Step 1. The consumer searches "wireless headphones" on Amazon (Mar 3). The rendered HTML is captured — every product shown is a Weitzman "box."

Pos	Product	Spons.	Rating	Price
1	Sony WH-10...	**Yes**	4.7	$298
2	Bose QC 45		4.5	$279
3	JBL Tune 760	**Yes**	4.3	$79
4	AirPods Max		4.6	$449
5	Sennheiser 450		4.4	$149
6	Beats Studio	**Yes**	4.2	$249
7	Anker Q30		4.5	$59
⋮	⋮		⋮	⋮
48	Skullcandy Evo		4.1	$99

48 products shown. Attributes (sponsored flag, rating, price) extracted from HTML.

From a search page to the data

Step 2. The clickstream records which products the consumer *opened*.

Pos	Product	Spons.	Rating	Price	**Open**
1	Sony WH-10...	**Yes**	4.7	$298	**1**
2	Bose QC 45		4.5	$279	**1**
3	JBL Tune 760	**Yes**	4.3	$79	0
4	AirPods Max		4.6	$449	**1**
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6	Beats Studio	**Yes**	4.2	$249	0
7	Anker Q30		4.5	$59	0
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48	Skullcandy Evo		4.1	$99	0

3 of 48 opened — the **products evaluated**. Non-opens are informative: z_j too low to justify opening.

From a search page to the data

Step 3. Amazon order history and URL detection identify the *purchase*.

Pos	Product	Spons.	Rating	Price	Open	**Purch.**
1	Sony WH-10...	**Yes**	4.7	$298	**1**	0
2	Bose QC 45		4.5	$279	**1**	**1**
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Cost per product = time on this intent ÷ products opened. Most intents end with *no* purchase.

Grouping searches into “intents”

An **intent** is a shopping goal that spans multiple searches, sessions, and merchants — not a single query.

Intent A = “headphones” (one participant, across days)

Date	Merchant	Keyword	Prod. seen	Prod. opened
Mar 3	Amazon	“wireless headphones”	48	3
Mar 3	Amazon	“bluetooth headphones”	52	5
Mar 5	Best Buy	“noise cancelling”	36	4
Mar 5	Amazon	“sony wh-1000xm5”	12	2

- Pages are tagged to the intent by matching their **HTML** (title, breadcrumb, product name) — *not* by session.
- Primary sample: **9,653** search-seeded intents; **≈85%** end in no purchase.

The clustering rule

Two queries join the same intent if TF-IDF cosine $\geq$ **0.3**, within a 14-day window (union-find: A–B and B–C $\Rightarrow$ A–B–C).

	wireless h.	bluetooth h.	noise canc.	sony wh-
wireless headphones	1.00	**0.71**	**0.52**	**0.34**
bluetooth headphones		1.00	**0.48**	**0.31**
noise cancelling h.			1.00	**0.38**
sony wh-1000xm5				1.00
cat food grain free	0.00	0.00	0.00	0.00
grain free cat food	0.00	0.00	0.00	0.00
cat food sensitive stom.	0.00	0.00	0.00	0.00
running shoes men	0.01	0.01	0.00	0.00

Bold ≥ 0.3: same intent. The four headphone queries collapse into one; cat food and shoes stay separate.

From clickstream to outcomes

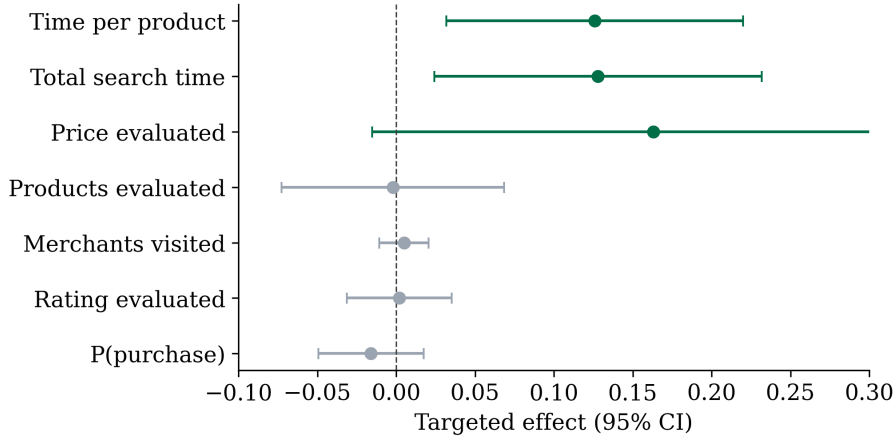
Unit of observation: participant i and intent (purchase goal) ℓ.

- **1) Classification:** each page visit $p \to \text{Intent}_{ip} \in \{\ell, \varnothing\}$.
- **2) Aggregation:** construct intent-level outcomes $Y_{i\ell}$ (e.g., time, price, purchase, ...).
- **3) Estimation (for each ℓ):**

$$Y_{i\ell} = \alpha_\ell + \beta_\ell \, \mathbf{1}\{\text{Targeted}_i\} + \gamma_\ell \, \mathbf{1}\{\text{Untargeted}_i\} + X_i' \delta_\ell + \varepsilon_{i\ell}.$$

SE clustered by participant.

Overview of the results: more effort, but no better outcomes



Green: more time, pricier products. Gray: same products, ratings, merchants, purchases.

More effort, no better outcome.

Additional specifications

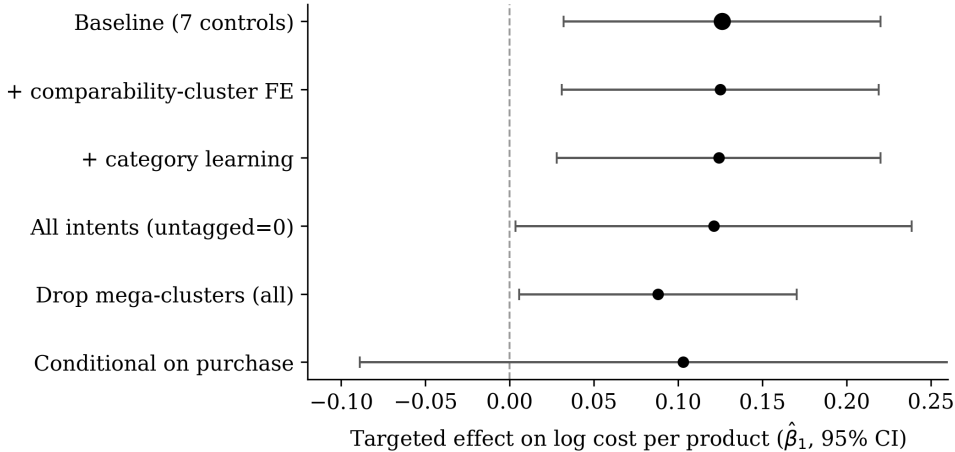
Outcome: **log time per product**. Coefficient on **Targeted** ($\hat{\beta}_1$), SE clustered by participant. Each row is a separate concern the effect could be an artifact of.

Specification	$\hat{\beta}_1$ (SE)	Concern it rules out
Baseline (7 demographic controls)	$+0.126^{***}$ (.048)	— (headline)
+ prior searches in same category	$+0.124^{**}$ (.049)	not just *learning*
+ product-category FE	$+0.120^{***}$ (.046)	compares *within category*

How to read each row.

- **Learning**: people who shopped a category before are faster; adding a prior-search count leaves $\hat{\beta}_1$ unchanged $\Rightarrow$ not learning. (Learning is real, -0.05^{***}, but *balanced across arms* — it's an RCT.)
- **Category FE**: compare Targeted vs No-Ads *within the same product category* $\Rightarrow$ like-with-like, not driven by a different basket of goods.

Robustness — coefficient plot



Every specification lands to the right of zero, near +0.12. Only **conditioning on purchase** widens to include zero — that is the bias.

Is the extra time informative — or friction?

The targeted arm spends +13% more time per product and per intent. Two interpretations with opposite welfare implications:

Informative

More time ⇒ a **sharper signal** about match quality ⇒ **better choice**.

Benchmark in Ursu, Wang & Chintagunta (2020): search time pays off.

vs.

Friction

More time ⇒ **same choice**; the extra time buys **nothing**.

Pure deadweight loss — our alternative.

Where κ comes from: the search model

Ursu, Wang & Chintagunta (2020) write a shopper's intent as a **three-stage** likelihood — and **time enters the last stage**:

$$\mathcal{L} = \prod_{\text{intents}} \underbrace{\Phi(z_j/\sigma_z)}_{\textbf{1. click}\text{: open product } j} \times \underbrace{f(\log t_j)}_{\textbf{2. dwell}\text{: time on } j} \times \underbrace{\frac{\exp\big(\mu_j/\sigma_\varepsilon(t_j)\big)}{\sum_k \exp\big(\mu_k/\sigma_\varepsilon(t_k)\big) + 1}}_{\textbf{3. buy}\text{: pick by value}}$$

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The whole question lives in how the value signal sharpens with dwell:

$$\sigma_\varepsilon(t_j) = \sigma_0\, t_j^{-\kappa} \quad \Rightarrow \quad \kappa > 0\textbf{: time informative} \quad \text{vs} \quad \kappa = 0\text{: time } \textbf{friction}$$

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Model B ($\kappa = 0$) is **nested** in Model A $\Rightarrow$ we estimate κ per arm and LR-test $\kappa = 0$ (df $= 1$).
μ_j = product value; t_j = dwell on j.

Estimating κ: informative without ads, not with them

Purchase-stage estimate on evaluated products, per arm. LR-test of $\kappa = 0$.

Arm	does time sharpen the choice?	LR test $\kappa = 0$
No Ads	**yes** — $\kappa > 0$	$p = 0.003$
Targeted	not distinguishable	$p = 0.49$ (NS)
Untargeted	not distinguishable	$p = 1.00$ (NS)

Estimating κ: informative without ads, not with them

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Without ads, dwell sharpens the value-based choice (**informative**). **With targeted ads**, time does **not** — a friction model cannot be rejected.

Contribution

1. **Method.** Measuring search cost over *all* intents (not just purchases) removes a selection bias that *hides* the effect of ads.
2. **Finding.** Personalized advertising *raises* search costs (+13% per product) with no offsetting benefit — it is friction, not information.
3. **Mechanism.** The extra time tracks *ad exposure* — more of the search budget is spent alongside advertising.
4. **Welfare.** A structural path to a dollar value of the deadweight loss.

Targeted ads are associated with *longer* search and no measurable gain in matches or purchases.

Thank you

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Backup

Backup — randomization balance

	No Ads (N=131)	Targeted (N=177)	Untargeted (N=155)	p (joint)
Age	50.15	53.15	50.92	0.152
Female	0.717	0.615	0.595	0.082
Non-white	0.205	0.224	0.235	0.828
College degree	0.685	0.690	0.601	0.186
Log income	10.91	10.78	10.82	0.435
Employed full-time	0.425	0.374	0.392	0.664
IT skills (1–4)	3.40	3.48	3.56	0.314

Balanced across arms. All regressions add these 7 controls and cluster SEs by participant.

Backup — a dollar value on the friction

Two distinct gradients. (i) The **estimated** time effect is *not* flat across income — it peaks in the **middle**. (ii) Each hour is then valued at the participant's **own** value of time (income/2080). Quality offset is not distinguishable from zero (κ NS), so the extra time is deadweight.

Income band	**Estimated effect** (targeted, time/product)	Value of time	Deadweight loss
Low ($15–39k)	+0.04 (NS)	$7/hr	$8 / yr
Mid ($62–87k)	+0.27***	$30/hr	$27 / yr
High ($124–175k)	+0.11 (p=.11)	$60/hr	$33 / yr

- Pure **deadweight loss**: same products / ratings / purchases, only more time.
- The **estimated friction is largest for the middle class** ($+0.27^{***}$) — not the rich. The dollar figure rises into the top band only because *their time is worth more*, not because the effect is.
- Value-of-time uses the full wage (income/2080); halving it (leisure < work) halves the dollar figures. The *sign* and *shape* are robust.